

QUESTION 01 / THE AUTUMN SALES JUMP

The revenue grew. Did baskets get bigger?

I began with monthly revenue, but that alone did not explain what changed. I compared distinct invoice counts and average order value before drawing a conclusion about the autumn increase.

6.07m

Recorded sales*

17,635

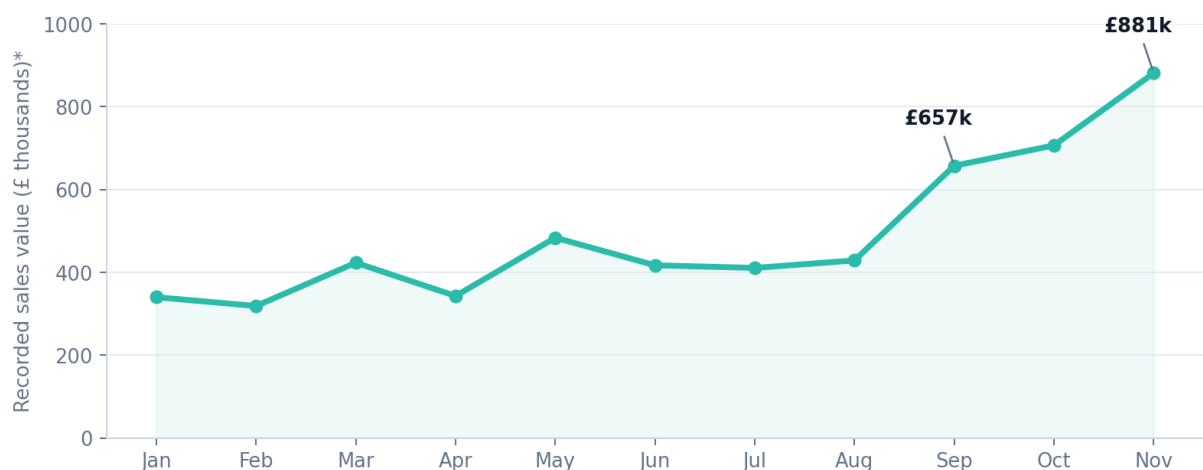
Distinct invoices

+34.1%

Sep–Nov sales

-12.4%

Sep–Nov AOV

Autumn sales increased, but the cause needs a second metric

September to November 2011: +34.1% recorded sales, +53.0% invoices, -12.4% average order value.

This is an arithmetic decomposition, not evidence about marketing campaigns, prices or causality. December 2011 ends on the ninth and must not be read as a full-month decline.

METHOD / THE DATE PROBLEM

The dates changed the story.

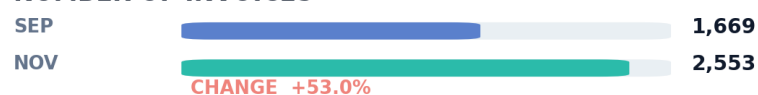
Before comparing months, I checked the first invoice dates. Invoice 536365 was recorded as 2010-01-12, but the expected sequence for this source begins on 1 December 2010. Month and day are reversed on rows where both components permit the swap.

Check	Observed result
Rows in supplied filtered file	384,721
Rows with day/month swapped	147,998
Invoice-date inversions in invoice-ID order	21 before repair / 0 after
Invoices with multiple corrected timestamps	30 (use earliest for invoice date)
Exact duplicate rows	5,178 (retained in main figures)
Exact-duplicate sensitivity to sales	23,119.37 (~0.38% of reported sales)

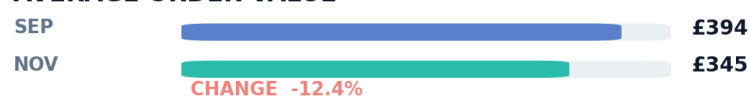
What the order-level check added

More invoices, not larger average orders

NUMBER OF INVOICES



AVERAGE ORDER VALUE



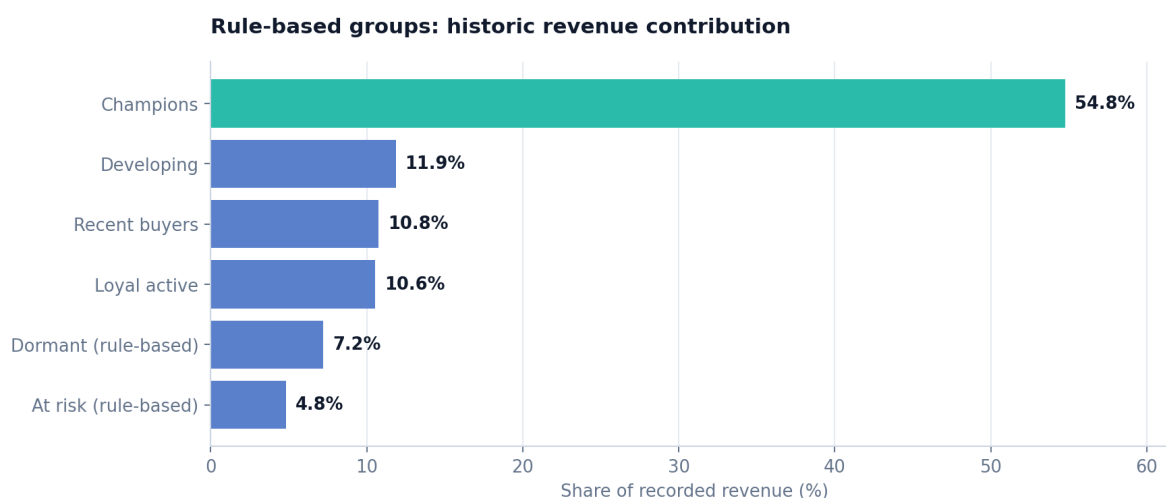
The growth in invoices is substantially larger than the growth in sales value because average order value falls over the same period. This does not identify which customer or marketing intervention caused the movement.

I kept duplicate rows in the headline figures because repeated identical line items can be legitimate; the separate sensitivity result shows how much removing them would change the total.

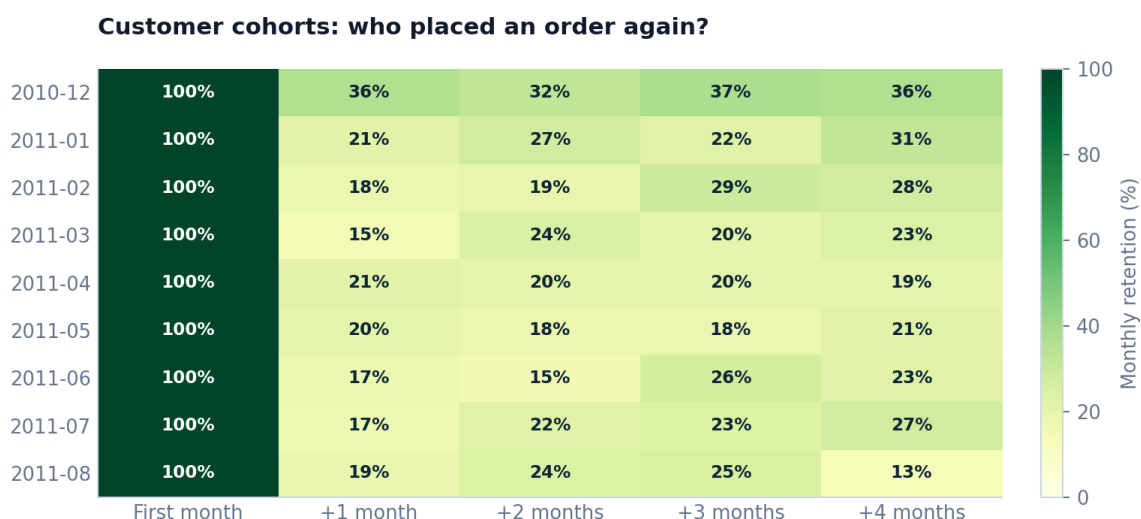
QUESTION 02 / CUSTOMER BEHAVIOUR

Returning is not the same as retention.

Of the 4,261 observed customers, 2,773 placed more than one invoice during the extract. This retrospectively defined group contributed 92.1% of recorded sales. The percentage sounds striking, but it cannot tell us what a new cohort will do next month.



Follow-up: first observed monthly cohorts

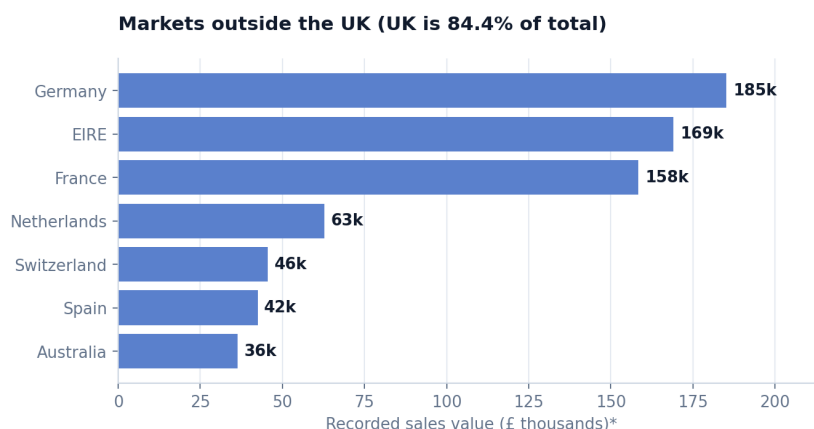


These rows are customers first observed in the respective month of this file, not confirmed first-ever customers. A cohort's month +1 rate is the share observed purchasing in that exact later month. RFM segment labels are hand-set descriptive rules, not a churn model.

BOUNDARIES / WHAT THIS FILE CAN ANSWER

Useful patterns. Clear limitations.

The United Kingdom accounts for 84.4% of observed sales. The international view is useful for describing this extract, but a geography comparison needs a warning: this is not a balanced, representative sample of global retail activity.



Checks I would keep before presenting this

Dates. The imported year, month, week, day and weekday were calculated from the inconsistent dates. I derived new calendar values after correcting ambiguous rows, then rechecked chronology against invoice identifiers.

Repeated items. Identical rows are not proof of a duplicate error. Keeping them leaves £6.07m recorded sales; dropping later exact copies would subtract £23,119.37 (~0.38%). Both outcomes are documented.

Population and timing. Only positive-quantity and positive-price transactions with known customer IDs appear here. Claims about refunds, true acquisition, profit, customer lifetime value or returns are not supported. The source covers Dec 2010 to 9 Dec 2011.

Currency. The extract has no currency field. Pound sterling is an explicit source-context assumption, not a fact encoded in the provided CSV.

If I extended the investigation, I would check whether the order-count increase was concentrated in specific first-observed cohorts, then compare that pattern by product mix. That requires more than one aggregate KPI.

Reproduction: the README documents the exact date correction, SQL queries, Python pipeline, an anonymized Power BI starter model, and data-lineage caveats. The provided filtered extract is not redistributed in the public-facing project archive.